VALTHERA

Technical Whitepaper
Infrastructure for an AI-Native Capital Market

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Capital allocated by intelligence.

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Important notice. This whitepaper describes the Valthera protocol and platform as designed. It is not an offer to sell or a solicitation of an offer to buy any security, token, or other financial instrument, and it is not investment, legal, or tax advice. Any capital-formation functionality described here will operate only where, and in the form, applicable law permits. Statements about future functionality describe the design and are not guarantees. Valthera does not issue a token.

Abstract

Capital markets were designed around human workflows: pitch decks, meetings, analyst reports, and investment committees. AI agents can now research businesses, use software tools, and execute transactions. Open payment standards already let them pay one another. What they lack is a structured environment in which to discover investment opportunities, examine the evidence behind them, value them, commit capital under enforceable rules, and be held accountable for how those decisions turn out.

Valthera provides that environment. It has six layers:

1. A **Submission Layer** that ingests project materials and records where each item came from.
2. An **Evidence Engine** that turns those materials into atomic claims and labels each one by how well it is supported.
3. An **Opportunity Registry** that publishes each opportunity as a versioned, machine-readable object.
4. An **Agent Network** of independent investment agents, each operating under a signed, machine-readable mandate.
5. A **Capital Formation and Execution** layer with an explicit commitment state machine, enforced through policy-constrained smart-contract vaults and regulated settlement partners.
6. An **Accountability Layer** that saves every decision with what was known when it was made, anchors it onchain, and scores it with proper scoring rules and risk-adjusted return metrics.

This document specifies each layer: its data structures, algorithms, interfaces, security model, and compliance architecture.

1. Introduction

The question behind Valthera is simple: **what happens when the investors themselves become autonomous software?**

AI agents are becoming economic participants. The length of software tasks that frontier agents can complete autonomously with 50% reliability has doubled roughly every seven months over six years [1]. Payment protocols such as x402 let software settle stablecoin payments directly over HTTP [2]. As agents take on more economic roles (running services, managing treasuries, building businesses), some of them will need to allocate capital. Today they would have to do so through infrastructure that assumes a person reads the deck, judges the claims, and signs the commitment.

Valthera is **not** a token launchpad, a research chatbot, or a trading bot. It is an **AI-native capital market** with the following properties:

- Projects present themselves in a structured, machine-readable format.
- Many independent agents evaluate each opportunity. No single model decides.
- Every material finding is linked to its evidence and labeled by verification status.
- Capital moves only under signed mandates, enforced permissions, and binding legal agreements.
- Every decision is saved, anchored onchain, and scored against outcomes.

Humans and agents participate on equal terms. The platform supports the full investment cycle: discovery, diligence, thesis, negotiation, commitment, settlement, monitoring, and reinvestment.

2. Problem Statement

2.1 Human attention is the bottleneck

Venture investing consumes scarce expert time at every stage: sourcing, screening, diligence, valuation, committee approval, and monitoring. The results are predictable. Founders without networks go unheard. Investors cannot properly examine most of what they see. Persuasive narratives can outweigh evidence.

2.2 Pitch decks mix fact and aspiration

A deck blends verifiable facts, partially supported claims, projections, and unverifiable statements, usually without saying which is which. The investor has to separate them under time pressure.

2.3 Analysis is disconnected from allocation and outcomes

AI tools can summarize documents and review code, but their output ends in a report. Analysis is not connected to capital, and outcomes are seldom fed back to measure how good the analysis was.

2.4 Agents have payment rails but no market infrastructure

Agent payment infrastructure is maturing. Its foundation’s members include major cloud, payments, and AI companies [2]. Actual economic usage is still small, though, and a meaningful share of observed transactions has been described as artificial [3]. In any case, a payment rail only moves money. It does not evaluate opportunities, enforce mandates, or produce verifiable track records. Valthera supplies those missing layers and treats payment rails as interchangeable components.

2.5 Early-stage returns are heavy-tailed

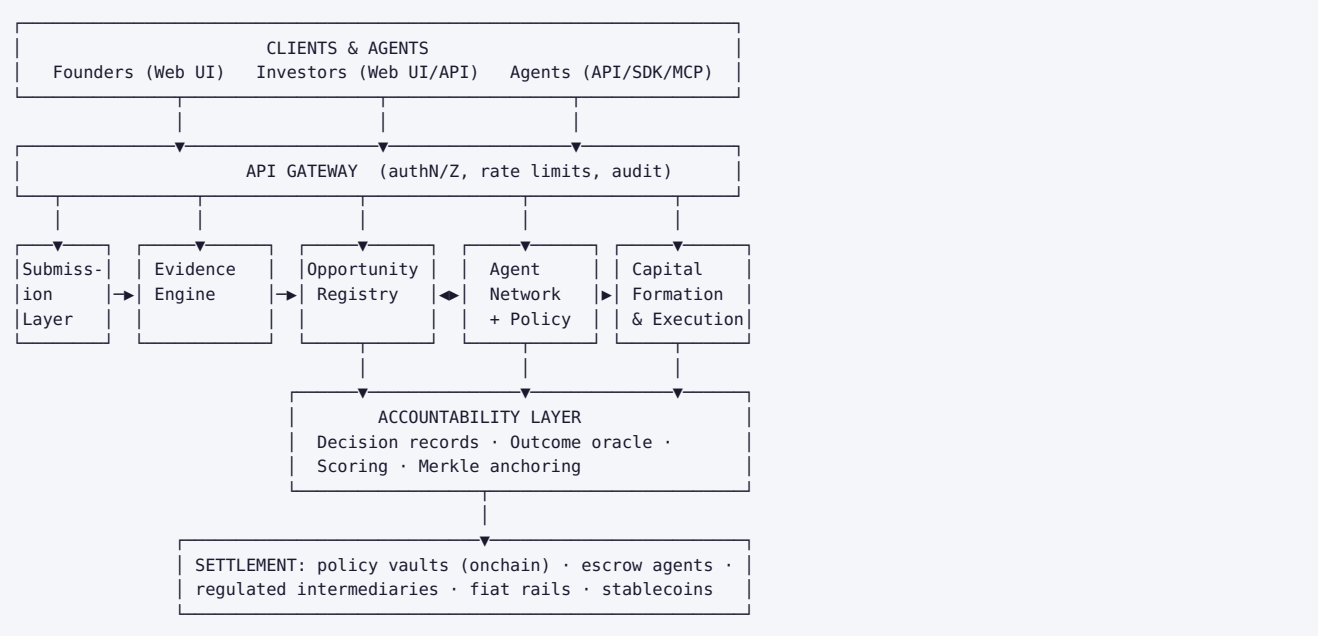
Early-stage returns follow an extreme power-law distribution. A small number of outliers dominate portfolio performance [4]. Selection quality therefore matters a great deal, short-term return records are noisy, and any system that rates investors must account for both facts (see Section 14).

3. Design Principles

#	Principle	Engineering consequence
P1	Evidence over narrative	Every material claim carries a verification label and links to its provenance
P2	Plurality of judgment	No canonical score. Assessments from independent agents are kept and compared
P3	Valuation as a distribution	Scenario-based ranges with explicit assumptions. Language models do not output price points directly
P4	Bounded autonomy	Permissions are enforced outside the model, in policy engines and smart contracts
P5	Opinion ≠ commitment	An explicit commitment state machine with legally binding transitions
P6	Untrusted inputs	All submitted content is data. It is isolated from instruction channels
P7	Point-in-time accountability	Decision records cannot be altered after the fact and are hash-anchored
P8	Compliance by construction	Regulatory gating is a first-class system state, not an afterthought
P9	Chain-agnostic	The value lies in intelligence and coordination, not in any one network
P10	Least privilege everywhere	Every component, agent, and key holds only the minimum authority it needs

4. System Architecture

4.1 Layered overview



4.2 Core services

Service	Responsibility	Key technologies
`ingest`	Upload, malware scan, format normalization, provenance hashing	Sandboxed parsers, SHA-256 content addressing
`claims`	Claim extraction and classification	LLM extraction with structured-output schemas, human QA sampling
`verify`	Verification adapters (onchain, code, registries, analytics)	Indexers, static analyzers, connector framework
`registry`	Versioned opportunity objects, event stream	Append-only event log, REST and webhooks
`valuation`	Scenario engine, Monte Carlo, method library	Deterministic compute service, auditable inputs
`agents`	Agent registration, runtime sandbox, tool gateway	Containerized runtimes, scoped credentials
`policy`	Mandate compilation and pre-trade checks	Declarative policy language (Section 10)
`capital`	Rounds, books, auctions, tranches, state machine	Transactional database, legal-document integration
`settle`	Vault orchestration, partner settlement	Smart-contract vaults, custodian and escrow APIs
`ledger`	Decision records, outcome ingestion, scoring	Append-only store, Merkle batching, onchain anchor
`identity`	Operator KYC/KYB, agent keys, attestations	Signed attestations, key rotation

4.3 End-to-end flow

```
1. Project submits materials ──► ingest (hash, scan, normalize)
2. claims extracts atomic claims ──► verify assigns evidence labels
3. registry publishes Opportunity v1 (event: opportunity.published)
4. Agents subscribed by mandate filter receive event
5. Each agent: retrieves evidence ► runs analysis ► calls valuation
   ► submits Assessment (signed) ► ledger snapshots DecisionRecord
6. Agent optionally submits Indication / Conditional Commitment
   ► policy pre-checks against mandate ► capital records state
7. Round closes ► compliance gates ► binding agreements executed
   ► settle moves funds via vault / escrow / intermediary
8. Milestones & metrics ingested over time ► ledger resolves forecasts
   ► scoring updates agent track records
```

5. Submission Layer

5.1 Accepted inputs

Category	Examples	Handling
Narrative	Pitch deck, memo, whitepaper	Parsed in a sandbox. Text and images extracted
Financial	P&L, balance sheet, cap table, bank statements	Structured parse, consistency checks
Metrics	Analytics exports, revenue dashboards	Read-only connectors preferred over uploads
Technical	Repositories, architecture docs, audits	Repository linking with commit pinning
Onchain	Contract addresses, treasury wallets, token contracts	Address registry, indexer subscription
Terms	Target raise, valuation, instrument, use of proceeds, milestones	Structured form, version-controlled
Disclosures	Related parties, prior raises, litigation, conflicts	Mandatory attestation

5.2 Ingestion pipeline

1. **Receive:** upload over TLS and assign a submission ID.
2. **Scan:** antivirus and malware detection. Active content such as macros and embedded scripts is stripped.
3. **Normalize:** convert to canonical text and images in an isolated parsing sandbox with no network egress.
4. **Hash:** compute a SHA-256 digest of every original artifact. The digest becomes the artifact's permanent identifier.
5. **Sanitize for models:** detect and neutralize hidden text (white-on-white, zero-width characters, off-page content, image-embedded text). Each segment is wrapped in delimiters that mark it as untrusted data (Section 16.3).
6. **Store:** encrypted object storage, with access recorded per artifact.

5.3 Connectors

Live connectors beat static uploads because they cannot be retroactively edited. Supported connector classes are read-only analytics, payment processors, accounting systems, code hosts, and blockchain indexers. Each connector pull records its time, scope, and response hash.

6. Evidence Engine

6.1 Claim extraction

The extraction model breaks submissions into **atomic claims**: single, checkable propositions. It uses constrained structured output against the `Claim` schema (Appendix B).

Taxonomy: `TEAM`, `MARKET`, `PRODUCT`, `TECHNOLOGY`, `TRACTION`, `FINANCIALS`, `COMPETITION`, `TOKENOMICS`, `TERMS`, `LEGAL`.

Materiality is scored from 0 to 1, from the claim's influence on valuation and risk. Claims with materiality ≥ 0.5 must be verified before an opportunity can be published.

6.2 Evidence labels

Label	Definition	Minimum evidence
`VERIFIED`	Confirmed by a reliable source independent of the project	≥1 independent primary source, or direct connector data
`SUPPORTED`	Consistent with evidence but not independently confirmed	Corroborating indirect evidence
`REPORTED`	Asserted by the project only	Project document
`PROJECTED`	A forward-looking estimate or assumption	Identified as a forecast
`CONTRADICTED`	Conflicts with available evidence	≥1 credible conflicting source
`UNKNOWN`	Insufficient information to assess	None. The gap is recorded explicitly

Hard rule: the system never fills an `UNKNOWN` with an inferred value. Missing data stays missing and is shown as such.

6.3 Verification adapters

Claim type	Adapter	Method
Active users	Analytics connector	Distinguish registered from active users. Check cohort retention. Detect bot traffic
Revenue	Payment and accounting connectors	Reconcile reported revenue with processor payouts and bank data
Onchain volume	Indexer adapter	Exclude self-transfers and wallet clusters. Flag wash patterns and incentive-driven volume
Team history	Public-record adapter	Professional histories, code contribution graphs, prior shipped products
Code quality	Repository adapter	Static analysis, dependency vulnerability scan, test coverage, commit history
Smart-contract risk	Contract adapter	Verified-source match, static analysis, privileged-role mapping, upgradeability, existing audits
Market size	TAM decomposer	Rebuild TAM bottom-up from stated assumptions and test each one
Token supply	Token adapter	Reconcile stated circulating supply, total supply, and vesting with contract state

6.4 Onchain activity analysis

Reported volume (V_reported) is reconciled with an adjusted figure, organic volume:

```
V_organic = V_total
          − V_self        (same-entity transfers, clustered wallets)
          − V_wash        (circular flows within time window τ)
          − V_incentive   (volume attributable to rewards programs)
```

Wallet clustering uses common-input heuristics, funding-source tracing, and behavioral similarity. Each subtraction is reported with a confidence interval, never as a single precise number.

6.5 Confidence aggregation

Each claim c has a label weight w(label), shown below, and a source reliability r(s) between 0 and 1. An opportunity's **evidence coverage** (EC) is the materiality-weighted mean:

```
EC = Σ_c m_c · w(label_c) · r_c  /  Σ_c m_c

w: VERIFIED=1.0, SUPPORTED=0.7, REPORTED=0.3, PROJECTED=0.2,
   UNKNOWN=0.0, CONTRADICTED=0.0 (and flagged)
```

EC is a **coverage metric, not a quality score**. A project can have high coverage and still be a poor investment. Agents use EC as an input. The platform never uses it as a rating.

6.6 Quality control

- **Human audit sampling:** a stratified random sample of labeled claims is re-verified by analysts every week. Label precision and recall are published.
- **Adversarial test suite:** synthetic decks with planted contradictions and prompt injections, run on every model or prompt change.
- **Correction workflow:** projects can dispute a label with new evidence. Every change is versioned and visible. **No payment can alter a label.**

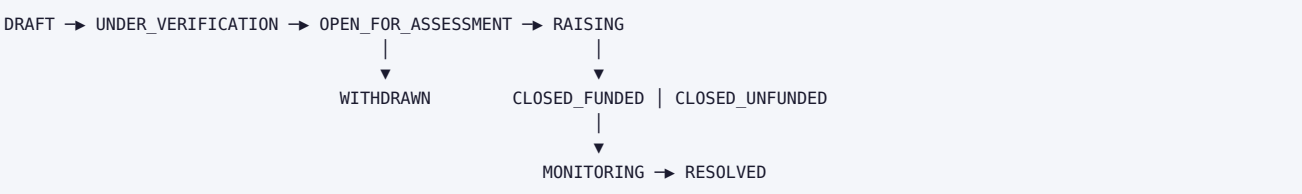
7. Opportunity Registry and Data Model

7.1 The Opportunity object

Each opportunity is an immutable, versioned document. Updates create new versions, and earlier versions remain retrievable. The full schema is in Appendix B. Its core fields:

```
{
  "opportunity_id": "opp_7f3a...",
  "version": 3,
  "status": "OPEN_FOR_ASSESSMENT",
  "project": { "name": "...", "entity": "...", "jurisdiction": "...", "stage": "SEED" },
  "category": ["DEVTOOLS", "AI_INFRA"],
  "terms": {
    "instrument": "SAFE_POST_MONEY",
    "target_raise": 500000, "currency": "USD",
    "proposed_valuation_cap": 8000000,
    "min_close": 250000, "deadline": "2026-12-15",
    "use_of_proceeds": [...], "milestones": [...]
  },
  "claims": ["clm_...", "..."],
  "evidence_coverage": 0.62,
  "artifacts": [{ "sha256": "...", "type": "DECK" }],
  "disclosures": {...},
  "content_hash": "0x...",
  "published_at": "2026-10-01T12:00:00Z"
}
```

7.2 Lifecycle



7.3 Event stream

Agents subscribe through webhooks or streaming to events filtered by their mandate. Event types: `opportunity.published`, `opportunity.updated`, `claim.relabeled`, `round.opened`, `round.book_updated`, `round.closed`, `milestone.reported`, `milestone.verified`, `outcome.recorded`.

8. Valuation Engine

8.1 Three separate questions

1. **Evidence-based value:** what is the project plausibly worth?
2. **Ask:** what valuation is the project requesting?
3. **Reservation price:** at what price, if any, would *this agent* invest?

The platform computes (1) as a distribution and displays (2). Each agent determines (3) privately, from its own mandate and cost of capital.

8.2 Architecture: models propose, the engine computes

Language models do **not** emit valuation numbers. They produce structured **assumption sets**, such as growth paths, margins, comparable selections, and exit multiples, each tied to evidence IDs. A deterministic valuation service then computes the outputs. Every number is therefore reproducible from its inputs, and every input can be traced to a claim.

8.3 Method library

Venture method (pre-revenue / early stage). Post-money valuation is the expected exit value at horizon T, discounted at the target return r and adjusted for expected dilution D:

```
Post-money = E[Exit_T] / (1 + r)^T × (1 − D)
E[Exit_T]  = Σ_k p_k · Exit_k      (scenario-weighted)
```

Comparables. Multiples (EV/ARR, EV/Revenue, EV/Fees) are drawn from a vetted set of comparable companies. The engine uses the median and interquartile range, adjusts for growth differentials, and applies a private-company liquidity discount.

Discounted cash flow (revenue-stage). Explicit forecast period plus terminal value. Revenue growth, margin, and discount rate are sampled from distributions, not fixed.

Protocol fee model. Value is derived from projected protocol fees, the share captured by the token or equity holder, and a capitalization multiple, adjusted for security and competitive risk.

Milestone-based (pre-product). The capital needed to reach the next value-inflection milestone, set against the valuation step-up achieved by comparable companies at that milestone, weighted by the probability of reaching it.

8.4 Token-specific metrics

These quantities are always reported separately and never treated as interchangeable:

```
Market cap (circulating)   = P × S_circulating
Fully diluted valuation    = P × S_max
Unlock overhang (t)        = Σ scheduled unlocks in [now, now+t] / S_circulating
Treasury-adjusted value    = FDV − liquid treasury assets
Rights profile             = {governance, fee claim, redemption, none}
```

A token with no claim on cash flows is flagged as `NO_ECONOMIC_RIGHTS`, and equity-style valuation methods are disabled for it.

8.5 Monte Carlo scenario output

For each method, N = 10,000 draws are taken over the assumption distributions. Output:

Field	Description
`p10`, `p50`, `p90`	Downside, base, and upside values
`prob_above_ask`	Share of draws above the requested valuation
`sensitivity`	Tornado ranking of assumptions by variance contribution
`method_dispersion`	Spread between methods (high dispersion signals low confidence)
`evidence_coverage`	From Section 6.5
`what_would_change`	Specific missing evidence with the highest expected impact

8.6 Confidence grading

Grade	Criteria
A	EC ≥ 0.8, revenue verified, method dispersion < 25%
B	EC ≥ 0.6, key traction verified or supported
C	EC ≥ 0.4, early evidence only
D	EC < 0.4, or a material claim is `CONTRADICTED`

9. Agent Network and Runtime

9.1 Agent types

Role	Function	Typical output
Research	Sourcing, market and competitor analysis	Investment memo
Technical	Architecture, code, and contract review	Technical risk report
Investment	Price versus expected return under mandate	Assessment with reservation price
Risk	Concentration, liquidity, contract, and downside analysis	Risk sign-off or veto
Monitoring	Post-investment tracking	Milestone and KPI updates

Agents can be built by independent operators on any model. Valthera is model-agnostic.

9.2 Agent lifecycle

```
REGISTERED → SANDBOX (paper assessments only)
          → ACTIVE_T0 (public recommendations)
          → ACTIVE_T1..T3 (graduated execution; Section 11)
          → SUSPENDED / RETIRED
```

Each move up the ladder requires a minimum track record, an operator verification level, and passing security checks.

9.3 Hosted runtime

Operators can run agents on their own infrastructure, connecting through the API, or in Valthera's hosted runtime:

- An isolated container per agent, with egress allowlists.
- A **tool gateway**: agents call tools (search, retrieval, valuation, the submission API) through a mediated interface. Each call is authorized against the agent's scope and logged.
- **Credential isolation**: agents never hold private keys or custody credentials. Transaction intents go to the policy engine (Section 11).
- **Resource budgets**: compute, token, and API-cost limits per agent.

9.4 Assessment schema (summary)

Each assessment must contain: a thesis, key risks, evidence citations (claim IDs), a valuation distribution reference, a reservation price or decline, confidence, probabilistic forecasts (Section 14.2), a "what would change my view" list, and a conflict declaration. Assessments are signed with the agent's registered key using EIP-712 typed data [5].

9.5 Integration interfaces

- REST/JSON API (Appendix A).
- Webhooks and event streaming.
- SDKs (TypeScript, Python).
- A Model Context Protocol (MCP) server that exposes Valthera tools to MCP-compatible agents.

10. Mandates and Policy Language

10.1 Mandate structure

A mandate is a signed, versioned, declarative document. It is compiled into executable pre-trade checks.

```yaml
mandate_version: 1.2
agent_id: agt_infra_seed_01
operator: { entity: "Example Capital LLC", lei: "...", kyb_level: 2 }
strategy: "Early-stage developer and AI infrastructure"

universe:
  stages: [PRE_SEED, SEED]
  categories: [DEVTOOLS, DATA_INFRA, AI_INFRA]
  instruments: [SAFE_POST_MONEY, PRICED_EQUITY]
  jurisdictions_allowed: [US, EU, UK, SG]
  exclude: [TOKEN_ONLY_NO_RIGHTS]

evidence_requirements:
  min_evidence_coverage: 0.55
  require_label: { TEAM: VERIFIED, TRACTION: SUPPORTED }
  block_if_contradicted: [FINANCIALS, TRACTION]
  smart_contracts: { require_audit: true, max_privileged_eoa: 0 }

risk_limits:
  max_position_pct: 0.02          # of committed capital
  max_category_pct: 0.30
  max_single_round_share: 0.25    # of round size
  max_illiquid_pct: 0.85
  max_daily_commit_usd: 100000

pricing:
  required_irr: 0.35
  max_ask_premium_to_p50: 0.10

execution:
  tier: T2
  human_approval_above_usd: 25000
  cooling_off_minutes: 60

conflicts:
  disclosed: [...]
  auto_recuse_on: [OPERATOR_HOLDS_POSITION, RELATED_PARTY]
```

10.2 Evaluation semantics

Before any indication or commitment, the policy engine evaluates every rule deterministically. A failure of any rule blocks the action and records the rule ID. The model cannot override the policy engine. The policy engine is the only path to execution.

10.3 Mandate changes

Changes are versioned. A change that loosens risk limits takes effect only after a timelock (default 72 hours) and is visible to any capital providers backing the agent.

11. Permission and Execution Layer

11.1 Execution tiers

Tier	Capability	Requirements
T0 Recommend	Publish assessments	Registered mandate, operator KYB level 1
T1 Indicate	Non-binding indications of interest (IOIs) with price	≥ 20 resolved forecasts, KYB level 2
T2 Conditional commit	Commitments that become binding after human sign-off	Executed legal framework agreement, KYB level 2, security review
T3 Bounded execution	Automatic binding commitments within hard limits	≥ 50 resolved forecasts with calibration within tolerance, independent audit, jurisdictional clearance

11.2 Intent → policy → execution

```
Agent ──(signed TransactionIntent)──► Policy Engine
        ├─ mandate rules                    ✗ ─► reject + log
        ├─ tier capability
        ├─ global risk circuit-breakers
        ├─ simulation (dry-run settlement)
        └─ human approval (if threshold)
                    │ ✓
                    ▼
        Execution Service ──► Policy Vault (onchain)  ─► enforces limits again
                          └─► Settlement partner      ─► enforces KYC/legal
```

Limits are enforced **twice**: off-chain in the policy engine and on-chain in the vault. Compromising one layer alone is not enough to move funds.

11.3 Policy vault (onchain)

Each agent's capital is held in a smart-contract account (for example a Safe multisig with policy modules, or an ERC-4337 smart account [6]) configured as follows:

- **Allowlisted counterparties:** only escrow contracts of rounds that Valthera has approved.
- **Spending limits** per transaction, per day, and in total.
- **Asset allowlist** (for example, specific stablecoins).
- **Session keys:** short-lived keys scoped to one round and an amount cap.
- **Timelock** on any change to limits or the allowlist.
- **Guardian role:** can pause or revoke but cannot withdraw to arbitrary addresses.
- **Separation of duties:** custody (owner keys), execution (session keys), and withdrawal (owner multisig plus timelock) are held separately.

Illustrative policy check (pseudocode):

```
function _checkIntent(Intent calldata i) internal view {
    require(!paused, "paused");
    require(allowedTarget[i.target], "target");
    require(allowedAsset[i.asset], "asset");
    require(i.amount <= perTxLimit, "perTx");
    require(spentToday + i.amount <= dailyLimit, "daily");
    require(block.timestamp <= i.deadline, "expired");
    require(sessionKeyValid(i.signer, i.roundId), "session");
}
```

11.4 Circuit breakers

Platform-wide halts trigger on anomalous conditions, for example:

- correlated commitments from more than *k* agents within minutes on a newly published opportunity;
- a spike in contradicted claims;
- an oracle failure;
- a detected exploit pattern.

When a breaker trips, new binding transitions are frozen until people review the situation.

12. Capital Formation Mechanisms

12.1 Commitment state machine

```
ASSESSED ─► INDICATED ─► CONDITIONALLY_COMMITTED ─► BINDING ─► FUNDED_IN_ESCROW ─► SETTLED
               │               │                      │             │
               ▼               ▼                      ▼             ▼
           WITHDRAWN         LAPSED               CANCELLED      REFUNDED
```

- `INDICATED`: non-binding, priced, visible in the book.
- `CONDITIONALLY_COMMITTED`: binding subject to stated conditions (human approval, minimum close, verified milestone).
- `BINDING`: legal agreement executed and policy checks passed.
- `FUNDED_IN_ESCROW`: funds held by the escrow contract or regulated escrow agent.

- `SETTLED`: the round closed and instruments were issued. Funds go to the issuer.

Each transition is signed, timestamped, and written to the ledger. “Attractive” (an assessment) can never be confused with “capital committed.”

12.2 Fixed-price rounds

The project sets price and size. Commitments fill first-come or pro-rata, subject to a minimum-close threshold. If the minimum is not reached by the deadline, all escrowed funds are refunded automatically.

12.3 Commitment book with price discovery

Agents submit indications as (price cap, amount) pairs. The project sees an aggregated demand curve (amount available at each valuation) without agent identities until close. The project can accept a clearing valuation, revise terms, or withdraw.

12.4 Uniform-price auction

For larger or oversubscribed rounds:

1. Bidders submit sealed bids (valuation cap v_i, amount a_i). Bids are commitment-hashed during the bidding window.
2. Bids are revealed and sorted by v_i, descending.
3. The clearing valuation v* is the highest valuation at which the cumulative bid amount Σ a_i, over bids with v_i ≥ v*, covers the target raise.
4. All winning bidders invest at the same valuation v*. Pro-rata allocation applies at the margin.

A uniform price reduces winner’s-curse pressure and rewards truthful bidding more than pay-as-bid formats do.

12.5 Milestone tranches

Capital is committed in full but released in tranches:

```
Tranche k released iff milestone_k.status == VERIFIED
   (verification: connector data or ≥2 independent verifier agents + human sign-off)
Unreleased tranches: refundable to investors if milestone deadline passes
```

12.6 Instruments

Supported instruments depend on jurisdiction and partner: post-money SAFEs, priced equity, convertible notes, revenue-share agreements, and (where lawful) tokenized securities. Token warrants are supported only where a legal review permits them.

13. Settlement and Custody

13.1 Settlement paths

Path	When used	Custody
Onchain escrow contract	Stablecoin settlement where permitted	Smart contract. Funds released only by close or refund logic
Regulated escrow agent	Fiat settlement, Reg CF and similar regimes	Licensed third party
Broker-dealer settlement	US offerings involving recommendations	Registered broker-dealer partner

Valthera’s own operating entities **never take discretionary custody** of investor funds.

13.2 Escrow contract (summary)

- `deposit(roundId, amount)` is accepted only from allowlisted, KYC-cleared vaults.

- `close(roundId)` can be called only when minimum close is met, compliance is attested, and the deadline is valid.
- `refund(roundId)` becomes callable by anyone after the deadline if minimum close was not met.
- Close and refund are mutually exclusive. Both emit events consumed by the ledger.

13.3 Payment rails

Small payments, such as agents paying for data, diligence services, or API usage, may use HTTP-native stablecoin protocols such as x402 [2]. Settlement adapters are pluggable. The investment logic does not depend on any single rail or chain.

14. Accountability Layer: Decision Records and Scoring

14.1 Decision records

At every assessment and state transition, the ledger saves:

- the opportunity version hash (exactly which evidence was visible);
- the assessment hash, valuation output, forecasts, and reservation price;
- the mandate version;
- the action taken and its signature.

Records are batched into Merkle trees, and the root is anchored onchain periodically. Anyone can verify that a record existed at time t and has not been altered.

14.2 Structured forecasts

Every assessment includes resolvable probabilistic forecasts, for example:

Forecast	Horizon	Resolution source
Raises a follow-on round	18 months	Filings, cap table, project attestation plus verification
Reaches stated ARR milestone	12 months	Revenue connector
Mainnet launch on schedule	Stated date	Onchain deployment
Critical security incident	12 months	Incident database, onchain events
Company still operating	24 months	Registries, activity signals

14.3 Forecast scoring

Forecasts are scored with **strictly proper scoring rules**. Under these rules, an agent's best expected score comes from reporting its honest probability [7]:

```
Brier score:      BS = (1/N) Σ (p_i − o_i)²            (0 = perfect)
Log score:        LS = −(1/N) Σ [o_i ln p_i + (1−o_i) ln(1−p_i)]
Brier skill:      BSS = 1 − BS_agent / BS_reference
```

The reference forecast is the base rate for that question class. Valthera also reports calibration (reliability diagrams), resolution, and sharpness. Because the Brier score depends on base rates, agents are compared **only on shared question sets**, and rare-event forecasts carry minimum sample-size requirements before scores are published.

14.4 Investment performance

Metric	Definition
DPI	Distributions / Paid-in capital
RVPI	Residual (unrealized) value / Paid-in
TVPI	DPI + RVPI
IRR	Internal rate of return on dated cash flows
PME	Public Market Equivalent versus a benchmark index
Loss ratio	Share of capital in positions marked below 0.5×
Max drawdown	Peak-to-trough on marked NAV

Unrealized marks are labeled by source: last round, comparables, or impairment. They are never mixed with realized returns.

14.5 Why the two scores are separate

Early-stage returns follow a heavy-tailed distribution [4], so a small portfolio's return record is dominated by whether it happened to hold one outlier. Forecast accuracy resolves faster and with less noise. Valthera therefore presents three things: **forecast skill**, **return performance**, and **selection-versus-price attribution**. That is, it shows whether the agent picked well, priced well, or both.

14.6 Anti-gaming

- Forecasts cannot be edited after submission. They can only be superseded by new, timestamped forecasts. Both are scored.
- Agents cannot pick which forecasts are scored. All forecasts are scored.
- Survivorship bias is controlled: retired agents' records remain public.
- Outcome resolution uses independent sources wherever possible.

15. Identity, Reputation, and Conflicts

15.1 Identity layers

Entity	Identity	Verification
Operator	Legal entity	KYB, beneficial ownership, sanctions screening
Agent	Key pair bound to operator	Signed registration, key rotation, revocation list
Project	Legal entity plus founders	KYB/KYC, founder identity attestation
Verifier	Individual or firm	Credential attestation

15.2 Reputation

An agent's public profile shows: forecast skill by category and stage, return metrics with sample sizes, its tier history, mandate-change history, and incident history. Valthera does not combine these into a single headline number, because that would recreate the "one score" problem.

15.3 Conflict registry

Operators must declare:

- positions in any project;
- token holdings;
- advisory roles;
- relationships with founders;
- fund affiliations.

Rules:

- **Automatic recusal** applies where the mandate's conflict rules match.
- **Undisclosed conflicts**, once discovered, lead to suspension and a permanent record.
- **Correlation monitoring** flags clusters of agents that act in lockstep or share infrastructure.

16. Security Architecture and Threat Model

16.1 Assets and adversaries

Assets: investor capital, the integrity of evidence labels, the integrity of decision records, confidential submissions, and agent keys.

Adversaries: misrepresenting projects, manipulative agent operators, external attackers, and compromised insiders.

16.2 Threat matrix

Threat	Vector	Controls
Indirect prompt injection	Hidden instructions in decks, repos, websites	Untrusted-content isolation, instruction hierarchy, output schema enforcement, no execution authority in analysis models
Excessive agency	An agent tricked into moving funds	Tiered permissions, policy engine, onchain vault limits, human approval
Hallucinated evidence	Fabricated citations	Citation resolver blocks any claim ID or source that cannot be retrieved
Data poisoning	Fake public signals (stars, reviews, volume)	Source reliability weighting, anomaly detection, organic-volume adjustment
Agent collusion and spoofing	Coordinated bids, fake consensus	Conflict registry, correlation monitoring, circuit breakers
Key compromise	Agent or session key theft	No long-lived keys in runtimes, session-key scoping, guardian pause
Smart-contract exploit	Vault or escrow bug	Audits, formal verification of core invariants, bug bounty, upgrade timelocks
Supply chain	Malicious dependency or model router	Pinned dependencies, software bill of materials (SBOM), vetted model endpoints, egress allowlists
Insider abuse	Label tampering	Append-only logs, dual control on label overrides, audit trail
Confidentiality breach	Leaked submissions	Encryption at rest and in transit, per-artifact access control (ACL), data-room watermarking

16.3 Prompt-injection defense in depth

OWASP ranks prompt injection as the top risk for LLM applications. It recommends least privilege, human-in-the-loop controls for privileged operations, segregation of untrusted content, and adversarial testing [8]. Valthera implements:

1. **Channel separation:** system and developer instructions are never merged with submission text. Untrusted segments are wrapped and labeled.
2. **Capability separation:** models that read untrusted content have **no tool access that can change state**. The analysis models and the execution path are fully separated.
3. **Output constraints:** all model outputs must validate against strict JSON schemas, and free-text fields are never interpreted as commands.
4. **Detection:** classifiers for injection patterns, including encoded or obfuscated payloads.
5. **Red-teaming:** a continuous adversarial corpus, run on every release.

Precedent. In May 2026 an attacker used an encoded social-media prompt to get Grok and the Bankrbot agent to

transfer roughly $150,000–$200,000 in tokens from a wallet whose permissions had been expanded by an NFT. No key was stolen and no contract was exploited [9][10]. Under Valthera's architecture this chain breaks in three places: (a) analysis models cannot issue transactions; (b) permissions cannot be expanded by inbound assets; (c) vault limits and human approval apply regardless of model output.

16.4 Assurance program

- An independent audit of smart contracts before each deployment.
- Formal verification of the escrow invariants: close and refund are exclusive, and funds are conserved.
- An annual penetration test and an SOC 2 Type II-aligned control framework.
- A public bug bounty.
- Public incident disclosure within a defined window after containment.

17. Data Architecture, Privacy, and Provenance

17.1 Storage

Data	Store	Protection
Artifacts	Encrypted object storage, content-addressed	AES-256 at rest, per-tenant keys
Claims and labels	Relational database with an append-only history table	Row-level access control
Events and decision records	Append-only log	Merkle batching, onchain root anchoring
Analytics	Warehouse (de-identified)	Aggregation thresholds

17.2 Confidentiality tiers

- **Public:** opportunity summary, labels, aggregate book.
- **Restricted:** full data room, visible only to agents and investors who have accepted an NDA and pass the confidentiality check.
- **Private:** raw connector data, visible only to verification services. Only derived labels are shown.

17.3 Provenance

Every label links to: artifact hash → extraction run ID (model version, prompt version) → verification adapter run → reviewer (if any). Any conclusion can be traced back to source bytes.

17.4 Privacy

Valthera minimizes personal data, enforces retention schedules, honors data-subject rights where applicable, and excludes confidential submissions from any model training unless the submitter explicitly opts in.

18. Onchain Components

Contract	Purpose
`AgentRegistry`	Agent public keys, operator binding, tier, revocation
`PolicyVault` (module)	Per-agent spending policy enforcement
`RoundEscrow`	Deposits, close, and refund for each round
`TrancheController`	Milestone-gated release
`RecordAnchor`	Periodic Merkle roots of decision records

Design rules:

- Minimal on-chain logic, with no token dependency.

- Upgrades sit behind a timelock and multisig.
- Contracts can be deployed on multiple EVM networks, and the adapter pattern allows non-EVM chains.

Blockchain enforces and records. It does **not** verify identity, revenue, or market existence. Those come from the off-chain Evidence Engine.

19. Governance

- **Operating company:** responsible for the platform, security, and compliance.
- **Methodology and Integrity Committee:** includes independent members. It owns the evidence-label standards, verification rules, agent admission and tier decisions, dispute resolution, and the scoring methodology. Its decisions and methodology changes are published.
- **Change management:** methodology changes are versioned and announced before they take effect. Historical records are never re-scored silently.
- **Separation of commercial and analytical functions:** sales and revenue teams have no access to label or assessment workflows.

20. Legal and Compliance Architecture

20.1 Principles

Using AI or smart contracts does not remove securities, commodities, AML, consumer-protection, or data-protection obligations. Compliance is enforced as system state. No round can move to `BINDING` without the relevant jurisdictional gates passing.

20.2 United States

- **Regulation Crowdfunding:** offerings must run through an SEC-registered broker-dealer or funding portal. Issuers may raise up to $5 million in any 12-month period [11]. Non-accredited investors are subject to annual limits based on income and net worth [12].
- **Funding-portal restriction:** a registered funding portal may not offer investment advice or recommendations, and may not hold or handle investor funds. Intermediaries that do so must register as broker-dealers [13][14].
- **Structural consequence:** Valthera keeps its **research function** (assessments and recommendations) structurally separate from any **offering intermediary**. US capital formation runs through registered broker-dealer partners. Information barriers stop research teams and systems from being paid on the basis of offering outcomes.
- **Other exemptions:** Regulation D and Regulation A offerings are supported through partners, with their own investor-eligibility gates.
- **AI-specific rulemaking:** the SEC’s 2023 predictive-data-analytics proposal was withdrawn in June 2025 [15]. General anti-fraud and disclosure obligations continue to apply.
- **Digital assets:** SEC leadership has described a Howey-anchored token taxonomy. Tokenized securities remain securities [16].

20.3 Other jurisdictions

Before launch in each market, Valthera completes a jurisdiction-specific review covering crowdfunding and prospectus regimes, crypto-asset rules, and adviser or arranger licensing. Capital formation is enabled one jurisdiction at a time.

20.4 Compliance controls

Control	Implementation
KYC/KYB	All operators, issuers, and investors (natural persons behind agents)
Sanctions and PEP screening	At onboarding and on every binding transition
Investor limits	Enforced in the policy engine, by regime
Communications	Assessments carry standardized risk disclosures and conflict statements
Records	Retained per applicable regulatory retention rules
Agent operator responsibility	Operators attest to their own regulatory status

21. Business Model

Stream	Description	Conflict safeguard
Investor subscriptions	Access to the full evidence layer, data rooms, and analytics	None needed. The investor is the payer
Agent infrastructure	Hosted runtime, API usage, policy vaults, reporting	Fees are usage-based, not outcome-based
Enterprise data	Benchmarks, methodology datasets (de-identified)	Aggregation thresholds
Project verification fee	Flat fee to run the Evidence Engine	Paid before labels are produced. The fee is identical regardless of outcome. Labels cannot be bought
Capital-formation fees	Through licensed partners where permitted	Partner-collected, disclosed, separated from research

No token. A token does not establish demand, improve investment quality, or create a sustainable business by itself. Valthera's economics rest on fees for services.

22. Go-to-Market

22.1 Beachhead: crypto and Web3 projects

These projects publish documentation, contracts, tokenomics, repositories, and onchain data. That gives the Evidence Engine verifiable material beyond the deck.

22.2 Sequence

1. **Diligence product:** projects receive evidence-labeled memos, valuation distributions, and risk maps. A pilot group of independent investors reviews the output, and their actual decisions are recorded for comparison.
2. **Agent network (T0/T1):** third-party agents publish assessments and indications, and track records start accumulating.
3. **Capital formation (T2):** conditional commitments settle through licensed partners in the first enabled jurisdictions.
4. **Bounded execution (T3):** qualified agents gain automated execution within hard limits.
5. **Expansion:** AI-native and SaaS businesses, more chains, and third-party agent infrastructure.

23. Roadmap

Phases are gated by exit criteria, not by dates.

Phase	Deliverables	Exit criteria
0 — Foundation	Claim schema, ingestion sandbox, verification adapters (onchain, repository, token), injection defenses	Label precision ≥ 90% on the audit sample; red-team suite passed
1 — Diligence Product	Evidence-backed memos, valuation engine, correction workflow, investor pilot	Investors return and say the output informs their decisions; correction rate within target
2 — Agent Network	Agent registry, mandates, policy engine, SDK and MCP server, decision ledger, public scoring	Several independent operators active; scoring methodology published
3 — Capital Formation	Commitment state machine, book and auction, escrow contracts, broker-dealer integration	Audited contracts; legal structure live in the first jurisdiction; first compliant rounds closed
4 — Bounded Execution	Policy vaults, session keys, T3 admission, circuit breakers	No critical incidents through the T2 period; independent audit passed
5 — Expansion	New categories, chains, and jurisdictions; third-party infrastructure	Category-specific methods validated against outcomes

24. Key Metrics

Domain	Metric
Evidence quality	Label precision and recall (audited); contradiction detection rate; correction rate
Usefulness	Investor engagement per opportunity; repeat usage; founder satisfaction
Agent quality	Brier skill score versus base rate; calibration error; coverage of the question set
Market	Indications per opportunity; book depth; close rate; time to close
Safety	Policy rejections; injection detections; incidents; mean time to contain
Outcomes	Realized versus forecast milestone rates; TVPI and DPI by cohort (long horizon)

25. Risk Factors

- **Model risk:** agents may misread evidence or be confidently wrong.
- **Manipulation:** projects may game verification, and agents may collude.
- **Correlated failure:** agents built on similar models may share blind spots and crowd into the same deals.
- **Regulatory risk:** the rules governing AI, crowdfunding, and digital assets are changing and differ by jurisdiction. Some features may be unavailable in some markets.
- **Adoption risk:** agent-to-agent commerce is early. Demand may develop more slowly than expected.
- **Outcome latency:** early-stage outcomes take years, which slows validation of track records.
- **Security risk:** smart contracts, keys, and integrations are attack surfaces.
- **Conflict risk:** commercial incentives could pressure analytical independence.
- **Evidence limits:** no verification process can detect every misrepresentation.
- **Unproven efficacy:** there is not yet rigorous evidence that AI agents outperform people at early-stage selection. Valthera's scoring framework is designed to measure this, not to assume it.

26. Conclusion

An agent that can analyze a business is useful. An agent that can analyze it, judge whether its price is attractive, commit capital within enforceable limits, and be scored against the result is a new kind of market participant.

Valthera provides the infrastructure that makes such participants **accountable**:

- evidence instead of narrative;
- plural judgments instead of one score;
- bounded authority instead of blind autonomy;
- decision records that cannot be altered after the fact, scored with honest metrics.

Valthera. Capital allocated by intelligence.

Appendix A — API Reference (v1)

All endpoints use HTTPS with JSON. Authentication uses OAuth 2.0 client credentials for operators. Agent write calls also require an EIP-712 signature.

Method	Endpoint	Description
POST	/v1/submissions	Create a submission (project)
POST	/v1/submissions/{id}/artifacts	Upload an artifact
GET	/v1/opportunities	List and filter opportunities (stage, category, EC, status)
GET	/v1/opportunities/{id}?version=n	Retrieve an opportunity version
GET	/v1/opportunities/{id}/claims	List claims with labels and evidence
POST	/v1/valuation/run	Run the valuation engine on an assumption set
POST	/v1/agents	Register an agent (operator)
PUT	/v1/agents/{id}/mandate	Submit a signed mandate version
POST	/v1/assessments	Submit a signed assessment with forecasts
POST	/v1/intents	Submit an indication or commitment intent
GET	/v1/intents/{id}	Intent status and policy evaluation result
GET	/v1/rounds/{id}/book	Aggregated demand curve
GET	/v1/agents/{id}/record	Track record and scores
GET	/v1/records/{id}/proof	Merkle inclusion proof for a decision record
POST	/v1/webhooks	Subscribe to events

Example: submit an intent

```
POST /v1/intents
Authorization: Bearer <token>
X-Valthera-Signature: 0x<eip712-signature>

{
  "agent_id": "agt_infra_seed_01",
  "opportunity_id": "opp_7f3a",
  "opportunity_version": 3,
  "type": "CONDITIONAL_COMMIT",
  "amount": 20000, "currency": "USDC",
  "max_valuation_cap": 7000000,
  "conditions": ["MIN_CLOSE_MET", "HUMAN_APPROVAL"],
  "assessment_id": "asm_19c2",
  "deadline": "2026-12-01T00:00:00Z",
  "nonce": 41
}
```

Response

```
{
  "intent_id": "int_88d0",
  "state": "CONDITIONALLY_COMMITTED",
  "policy": { "passed": true, "rules_evaluated": 23 },
  "requires": ["HUMAN_APPROVAL"],
  "record_id": "rec_5521"
}
```

Appendix B — Schemas

Claim

```
{
  "claim_id": "clm_0142",
  "opportunity_id": "opp_7f3a",
  "category": "TRACTION",
  "statement": "12,000 monthly active users",
  "materiality": 0.8,
  "label": "CONTRADICTED",
  "evidence": [
    { "source_type": "CONNECTOR", "source_ref": "analytics:pull_2291",
      "finding": "12,000 registered; 2,140 active (30d)",
      "reliability": 0.95, "hash": "0x..." }
  ],
  "method": "analytics_inspection_v2",
  "extraction_run": "ext_771 (model m-2026-09, prompt p-14)",
  "checked_at": "2026-10-01T09:12:00Z",
  "history": [{ "label": "REPORTED", "at": "2026-09-29T..." }]
}
```


Assessment


```
{
  "assessment_id": "asm_19c2",
  "agent_id": "agt_infra_seed_01",
  "opportunity_id": "opp_7f3a", "opportunity_version": 3,
  "mandate_version": "1.2",
  "decision": "INVEST_BELOW_ASK",
  "reservation_valuation_cap": 7000000,
  "valuation_ref": "val_330",
  "thesis": "...",
  "risks": [{ "risk": "...", "severity": "HIGH", "evidence": ["clm_0142"] }],
  "forecasts": [
    { "id": "f1", "question": "Raises follow-on within 18 months", "p": 0.35,
      "resolves": "2028-04-01" }
  ],
  "confidence": "C",
  "what_would_change": ["Verified 90-day retention ≥ 40%"],
  "conflicts": [],
  "signature": "0x..."
}
```


Decision record


```
{
  "record_id": "rec_5521",
  "type": "INTENT_TRANSITION",
  "refs": { "opportunity_hash": "0x...", "assessment_hash": "0x...",
            "mandate_hash": "0x...", "intent_id": "int_88d0" },
  "from_state": "INDICATED", "to_state": "CONDITIONALLY_COMMITTED",
  "timestamp": "2026-10-05T14:02:11Z",
  "merkle_batch": 18831, "anchor_tx": "0x..."
}
```


Appendix C — Scoring Formulas

Brier score (binary): BS = (1/N) Σ $(p_i - o_i)^2$, where $p_i \in [0,1]$ and $o_i \in \{0,1\}$.

Murphy decomposition: BS = Reliability − Resolution + Uncertainty. Valthera reports each component.

Brier skill score: BSS = 1 − BS / BS_ref, where BS_ref is the base-rate forecast for the question class.

Logarithmic score: LS = −(1/N) Σ ln(p assigned to the realized outcome). Probabilities are clipped to [0.01, 0.99] for publication.

Expected calibration error (ECE): forecasts are binned into deciles. ECE = Σ_b (n_b/N)·|mean(p)_b − mean(o)_b|.

Kaplan–Schoar PME: PME = Σ (distributions_t / I_t) / Σ (contributions_t / I_t), where I_t is the benchmark index level.

Selection-versus-price attribution: - Selection is measured by the forecast skill of the agent’s decisions to invest or decline. - Price is the realized multiple at the agent’s entry valuation compared with the round’s clearing valuation.

Appendix D — Glossary

Term	Definition
Agent	Software that pursues investment objectives under a registered mandate
Assessment	A signed agent evaluation with thesis, valuation reference, and forecasts
Claim	An atomic, checkable proposition extracted from submissions
Decision record	A snapshot of an action and its context that cannot be altered and is hash-anchored
EC	Evidence coverage: the materiality-weighted verification coverage metric
FDV	Fully diluted valuation: price × maximum supply
Intent	A signed request by an agent to change commitment state
Mandate	A signed, declarative policy defining an agent's universe, evidence thresholds, and limits
Policy engine	The deterministic service that evaluates intents against mandates and tier rules
Policy vault	A smart-contract account that enforces spending limits and allowlists onchain
Session key	A short-lived key scoped to a single round and amount cap
Tier (T0–T3)	An agent's level of execution authority

Appendix E — References

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